

Cristoph Paar: Lecture 2: Modular Arithmetic & Historical Ciphers.

- * Goal: computation in finite sets
- * the mod / "modulus" operator - like the remainder operator

* Let $a, r, m \in \mathbb{Z}$, and $m > 0$,
we write,

$$a \equiv r \pmod{m}$$

(if m divides $(a-r)$, i.e. $m \mid (a-r)$)

this is read as
" m divides $(a-r)$ "

* examples $\Rightarrow$

$$\begin{aligned} a &= 13 & 13 &= 4 \pmod{9} \\ m &= 9 & \Rightarrow & a - r = (13 - 4) = 9 \\ r &= ? \end{aligned}$$

* computation of the remainder,

* given: $a, m \in \mathbb{Z} \Rightarrow a = qm + r$

quotient $\swarrow$
remainder $\nwarrow$

* example $\Rightarrow$

$$\begin{aligned} a &= 42 & \Rightarrow & 42 = q \times 9 + r & \Rightarrow & \begin{aligned} q &= 4 \\ r &= 6 \end{aligned} \\ m &= 9 \end{aligned}$$

but also,

$$42 = 3 \times 9 + 15 \Rightarrow r = 15$$

$$\text{check} \Rightarrow (42 - 15) = 27, 9 \mid 27 \checkmark$$

but also

$$42 = 5 \times 9 + (-3) \Rightarrow r = -3$$

$$\text{check} \Rightarrow (42 - (-3)) = 45, 9 \mid 45 \checkmark$$

$$\begin{aligned} \text{check} &\Rightarrow 42 - 6 = 36 \\ &9 \mid 36 \checkmark \end{aligned}$$

$\therefore$ the remainder
is always not
unique.

* Equivalence Classes

* Examples $\Rightarrow$ when $a = 12, m = 5,$

$$12 \equiv 2 \pmod{5} \Rightarrow \text{check } 5 \mid (12-2) \checkmark$$

$$12 \equiv 7 \pmod{5} \Rightarrow \text{check } 5 \mid (12-7) \checkmark$$

$$12 \equiv -3 \pmod{5} \Rightarrow \text{check } 5 \mid (12-(-3)) \checkmark$$

so, the set $\{ \dots, -8, -3, 2, 7, 12, 17, \dots \}$

forms an "equivalence class" for modulo 5. All members of the class behave equivalently to modulo 5.

* Lets look at all equivalence classes for modulo 5,

$$\{ \dots, -10, -5, 0, 5, 10, \dots \} - A$$

$$\{ \dots, -9, -4, 1, 6, 11, \dots \} - B$$

$$\{ \dots, -8, -3, 2, 7, 12, \dots \} - C$$

$$\{ \dots, -7, -2, 3, 8, 13, \dots \} - D$$

$$\{ \dots, -6, -1, 4, 9, 14, \dots \} - E$$

from these two lists

$$D = B - D$$

* Lets solve $\underline{13 \times 16 - 8} = 208 - 8 = 200 \equiv 0 \pmod{5}$

* Similarly, $\underline{3 \times 1 - 3} = 3 - 3 = 0 \pmod{5}$ mod 5,
or steps of 5

$$8 \times 6 - (-7) = 48 + 7 = 55 \equiv 0 \pmod{5}$$

* Important Application

* $E \Rightarrow 3^8 \pmod{7} \equiv ?$

$\leftarrow$ exponentiation is what happens in most of cryptography. but the numbers are very large

* 1st method $\Rightarrow 3^8 = 6561 = 2 \pmod{7}$

* 2nd method $\Rightarrow 3^8 = 3^4 \times 3^4 = 81 \times 81$

equivalent of picking from
an equivalent class $\longrightarrow$

$$\begin{array}{ccc} 81 & \downarrow & \text{mod } 7 \\ 4 & \times & 4 = 16 \end{array}$$

basically the
shifty around
we did
before

* Rings: An algebraic view (on modular arithmetic)

* definition $\Rightarrow$ the "integer ring" $\mathbb{Z}_m$ consists of

1. The set $\mathbb{Z}_m = \{0, 1, \dots, m-1\}$

2. Two operators "+" and "*", such that for all $a, b, c \in \mathbb{Z}_m$,

$$(i) \quad a + b = c \pmod{m}$$

$$(ii) \quad a \cdot b = d \pmod{m}$$

* we can add, multiply any two numbers and the result is always in the ring. A ring is said to be closed.

* addition and multiplication are associative,

$$a + (b + c) = (a + b) + c$$

$$a \cdot (b \cdot c) = (a \cdot b) \cdot c$$

* there is the neutral element 0 with respect to addition, i.e., for every element $a \in \mathbb{Z}_m$, it holds $a + 0 = a \pmod{m}$.

* for any element 'a' in the ring, there is always the negative element, such that $a + (-a) = 0 \pmod{m}$.

(additive inverse always exists).

* there is a neutral element 1 with respect to multiplication, for every element $a \in \mathbb{Z}_m$, it holds $a \times 1 = a \pmod{m}$.

- the multiplication inverse exists only for some, but not for all elements. Let $a \in \mathbb{Z}$, the inverse a^{-1} is defined such that

$$a \cdot a^{-1} = 1 \pmod{m}$$

if inverse exists for a , we can divide by this element
since $\boxed{b \mid a = b \cdot a^{-1} \pmod{m}}$

- for multiplicative inverse,

let $m=9$, $a=2$, $a^{-1}=?$, $\mathbb{Z}_9 = \{0, 1, 2, 3, 4, \dots\}$

$$\Rightarrow 2 \cdot 2^{-1} = 1 \pmod{9}$$

but $2^{-1} = \frac{1}{2} = 0.5$, is not an integer, $\notin \mathbb{Z}_m$,

so, we are looking for

$$2 \cdot _ = 1 \pmod{9}$$

$$2 \times 5 \equiv 1 \pmod{9}$$

$$2^{-1} \equiv 5 \pmod{9}$$

$$6 \times _ \equiv 1 \pmod{9}$$

$$\gcd(6, 9) = 3 \neq 1$$

$\therefore$ the inverse does not exist

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